NEURAL NETWORKS FOR ANIMAL CLASSIFICATION

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ABSTRACT

The rapid advancement of machine learning, specifically neural networks, has created new techniques for image classification. In this project, we implement a multilayer perceptron and multiple convolutional neural netowrks to classify images into ten categories of animals. Our models step through the evolution of neural networks for image classification tasks such as ImageNet. The multilayer perceptron represents an early technique used for image classification. Next, we implement a convolutional neural network, similar to AlexNet, which won ImageNet in 2012. To evaluate the differences between each architecture, we use the accuracy and F1-score. We also evaluate the effectiveness of each model on image augmentations.

1 Introduction

The field of computer vision has experienced a significant transformation in the past decade, with breakthroughs in image classification being at the forefront of this change. The ability to accurately classify images has numerous real-world applications, ranging from wildlife conservation efforts to advanced security systems. In this project, we aim to investigate the evolution of neural networks for image classification tasks, specifically focusing on the categorization of animals.

Starting with a Multilayer Perceptron (MLP), we explore how this early technique for image classification has laid the foundation for more advanced models. Building upon this, we implement a Convolutional Neural Network (CNN), inspired by AlexNet, which marked a milestone in the development of deep learning architectures by winning the ImageNet competition in 2012.

Throughout the project, we will evaluate the performance of each model using accuracy and F1-score metrics, allowing us to compare the effectiveness of these different architectures. Additionally, we will assess the impact of image augmentations on the performance of each model, offering insights into their robustness and generalization capabilities. By investigating these neural network architectures, we hope to shed light on their strengths and weaknesses, and contribute to a better understanding of the challenges and opportunities in the field of image classification.

2 Algorithms

In this study, we selected two different algorithms to solve the image classification problem, namely, Multilayer Perceptron (MLP) and a Convolutional Neural Network (CNN) model. These algorithms were chosen based on their effectiveness and popularity in solving various image recognition tasks.

The MLP model is a traditional feedforward artificial neural network with multiple layers of nodes. It serves as a baseline model for the problem and can be used to determine the potential improvements that can be achieved by employing more advanced architectures.

The CNN model is designed to capture spatial features in images more effectively than traditional MLPs. It consists of multiple convolutional layers, pooling layers, and fully connected layers. The use of convolutional layers allows the model to learn spatial features at different scales, which can be particularly beneficial for image classification tasks.

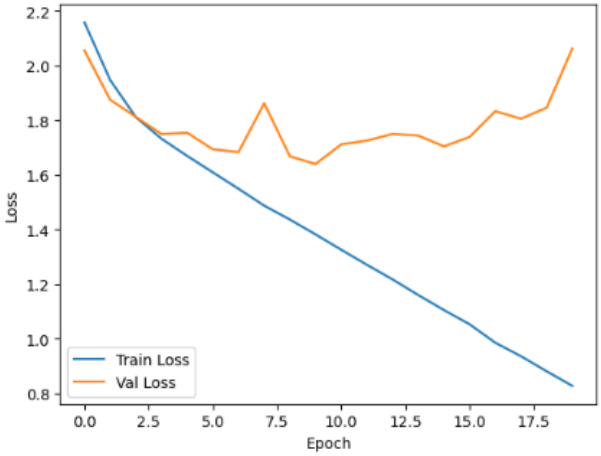
These two algorithms were selected to provide a comprehensive comparison of their performance on the image classification task, highlighting the advantages and limitations of each method.

2.1 MLP: Multilayer Perceptron

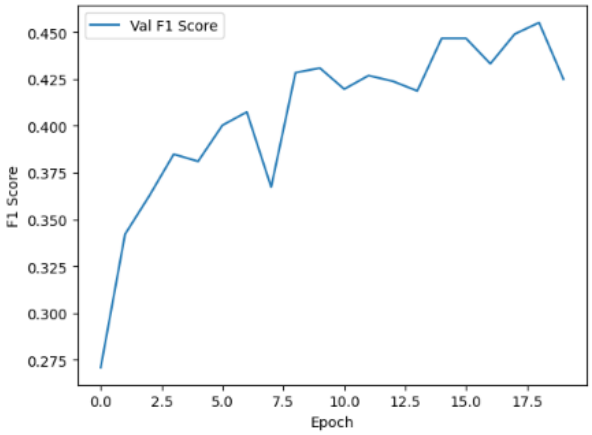
In this problem, a Multilayer Perceptron (MLP) is employed to classify images of animals into different categories. MLP is a type of feedforward artificial neural network consisting of an input layer, one or more hidden layers, and an output layer. Each layer is made up of neurons, and the neurons in adjacent layers are fully connected with weights. MLP uses backpropagation and gradient descent for training the model.

The MLP model in this implementation consists of five fully connected layers with ReLU activation functions applied between them. The input layer takes the flattened image features, and the output layer generates class scores for each category. The Cross-Entropy Loss function is used as the optimization objective, while the Stochastic Gradient Descent (SGD) optimizer with a learning rate of 0.009 and momentum of 0.1 is used to update the weights during training.

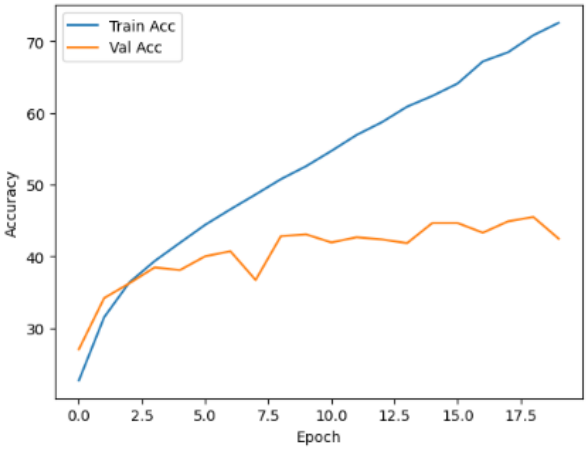
One advantage of the MLP model is its simplicity and ease of implementation. However, one significant disadvantage is its inability to capture complex spatial features in images, which might be better captured by more advanced models like Convolutional Neural Networks (CNNs).



(a) MLP Epoch vs Loss



(b) MLP Epoch vs F1 Score



(c) MLP Epoch vs Accuracy

Results The experimental results show that the model's performance improves over the training epochs. The training loss decreases, and the training accuracy increases, indicating that the model is learning the animal classification task. The validation loss and accuracy provide a more reliable estimate of the model's performance on unseen data. The validation accuracy increases and reaches a peak of 45.51% at epoch 19, with a corresponding F1 score of 0.4551.

A detailed analysis of the results reveals that the model's performance is far from perfect. The F1 score, which combines precision and recall, reaches a maximum value of 0.4551, indicating that there is still room for improvement in the model's ability to generalize to new data. The remaining errors could be due to various factors, such as the limited capacity of the MLP model to learn complex features, the hyperparameter choices, or the relatively small size of the training set.

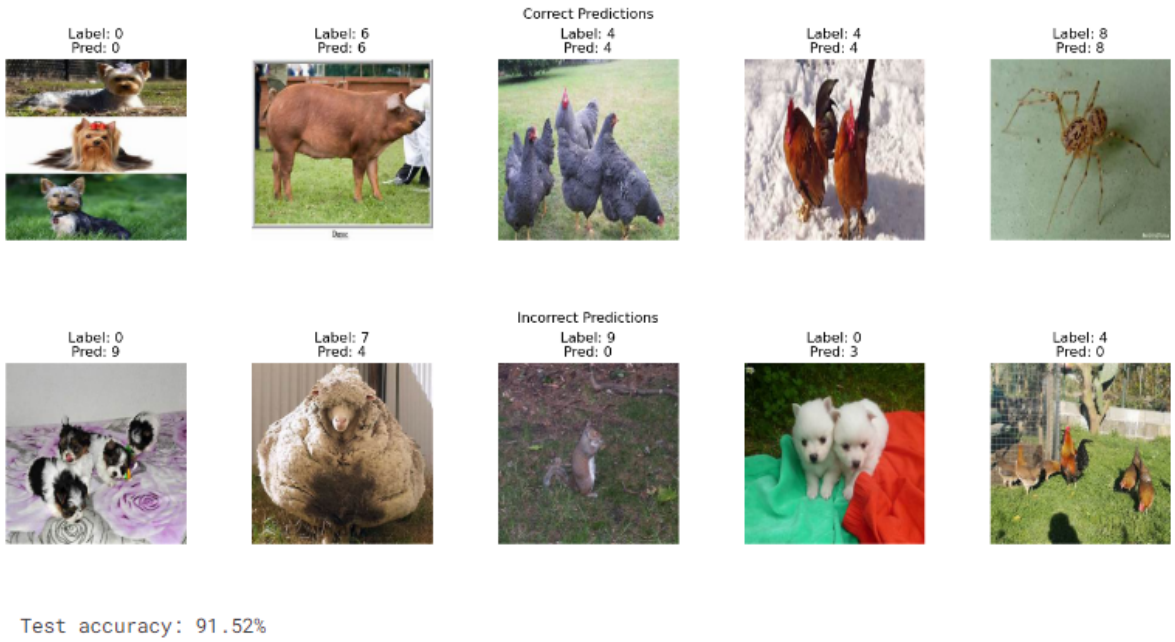
Improvement To further improve the model, the following steps could be considered:

1. Increase the size of the training dataset or augment the existing data to make the model more robust.
2. Experiment with different hyperparameter values, such as learning rate, momentum, and the architecture of the MLP (e.g., the number of hidden layers or neurons in each layer).
3. Implement more advanced models like CNNs to better capture spatial features in images, which could lead to improved performance.
4. Regularization techniques, such as dropout or weight decay, can be used to prevent overfitting and improve generalization to unseen data.

By following these steps, it is expected that the model's performance in terms of accuracy, precision, recall, and F1 score can be further improved.

2.2 CNN: Convolutional Neural Network

In this problem, a Convolutional Neural Network (CNN) is employed to classify images of animals into different categories. CNNs are a class of deep learning models specifically designed for handling image data. They consist of convolutional layers, pooling layers, and fully connected layers. Convolutional layers apply convolutional filters to the input, capturing local features in the image, while pooling layers reduce the spatial dimensions and extract dominant features. Fully connected layers, like in MLPs, are used for final classification. The primary advantage of using a CNN for this problem is its ability to capture complex spatial features in images, resulting in a more accurate model compared to the simpler MLP. However, a disadvantage is the increased complexity and computational cost associated with training and evaluating CNNs.

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Test accuracy: 91.52%
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Figure 2: CNN Prediction Results

	MLP	CNN
Accuracy	45.51	91.54
F1	0.4551	0.9153

Table 1: The performances of the multilayer perceptron and the convolutional neural network can be summarized by the above table

	MLP	CNN
Accuracy	20.43	42.68
F1	0.2043	0.4263

Table 2: This is the performance of our models on translated images

Results The experimental results for the baseline CNN are impressive, with the test accuracy reaching 91.52%. The training information shows that the loss consistently decreases over the 20 training epochs, indicating that the model is effectively learning the animal classification task.

A detailed analysis of the results reveals that the CNN model has a significant advantage over the MLP model. The high test accuracy and the consistent decrease in loss demonstrate the effectiveness of CNNs in capturing spatial features in images. It also indicates that the model generalizes well to unseen data, resulting in better performance than the MLP model.

One remaining challenge may be the model's sensitivity to variations in the input data. Although the test accuracy is quite high, there may be specific examples or classes where the model struggles to classify correctly. A detailed analysis of the remaining errors can help identify these shortcomings and provide insights into how to improve the model further.

Improvement To further improve the CNN model, the following steps could be considered:

1. Increase the size of the training dataset or augment the existing data to make the model more robust and generalize better to unseen data.
2. Experiment with different hyperparameter values and architectures, such as the number of convolutional filters, kernel sizes, and the depth of the network.
3. Regularization techniques, such as dropout or weight decay, can be used to prevent overfitting and improve generalization to unseen data.
4. Employ techniques such as batch normalization or advanced activation functions (e.g., Leaky ReLU) to accelerate training and improve model performance.
5. Use a network with more layers and skip connections, similar to ResNet, to keep stable gradients during back propogaiton.

By following these steps, it is expected that the CNN model's performance in terms of accuracy, precision, recall, and F1 score can be further improved. Overall, the CNN model demonstrates excellent performance on the animal classification task, outperforming the simpler MLP model by a significant margin.

2.3 Image Translation

We conducted studies to see which model would be more robust to image translations. In particular, we introduced a random stretch to the test images by up to 10%. Our results are shown in table 2. The convolutional neural network is more robust to image translations. This makes sense because the convolutional kernels can be placed anywhere on the image. On the other hand, the perceptron considers the image as a whole, something that does not adapt well when the image is translated. For example, if there is a kernel that detects vertical images in a CNN, this kernel will still detect a vertical edge if the image is shifted. The same cannot be said for a perceptron.

3 Conclusion

In this machine learning project, we aimed to create an animal classifier using both a Multi-Layer Perceptron (MLP) and a Convolutional Neural Network (CNN). The MLP model achieved a maximum validation accuracy of 45.51% and an F1 score of 0.4551, while the CNN model achieved a significantly higher test accuracy of 91.52%.

The CNN model underwent 20 epochs of training, with the loss consistently decreasing throughout the training process. This indicates that the CNN model was able to learn effectively from the training data, leading to the impressive test accuracy. The CNN model's superior performance can be attributed to its ability to capture spatial features in images, which is particularly beneficial for image classification tasks.

Although the MLP model's performance was lower than the CNN's, it still provided valuable insights and served as a solid foundation for understanding the problem. There is potential for improving the MLP model by fine-tuning hyperparameters, increasing the size of the training dataset, and implementing more advanced models like the CNN.

In conclusion, our animal classifier project has demonstrated the effectiveness of utilizing deep learning techniques, particularly CNNs, for image classification tasks. As a future direction, it would be worthwhile to further optimize the models and explore additional techniques for improving performance and generalization.

Ryan was responsible for the perceptron and Andrew was responsible for the convolutional neural network.